CHAPTER 11

THERMODYNAMICS

Key idea: heat supplied = increase in internal energy + work done by the gas.

1 FIRST LAW

$$\Delta U = Q - W$$

Symbol	+ when	− when
Q	heat into gas	heat out
W	gas expands	gas compressed
ΔU	T rises	T falls

2 INTERNAL ENERGY

$$\Delta U = nC_V\Delta T = \tfrac{f}{2}nR\Delta T$$

- Ideal gas: random KE only, so U depends on T only
- True for **any** process between the same temperatures
- Closed cycle: $\Delta U = 0$

3 WORK DONE BY A GAS

$$W = \int p\,dV \qquad W = p\Delta V$$

- W = **area** under the p–V graph
- Constant pressure: $W = p\Delta V$
- Cycle: net work = area **enclosed** (clockwise: done by gas)

4 MOLAR HEAT CAPACITIES

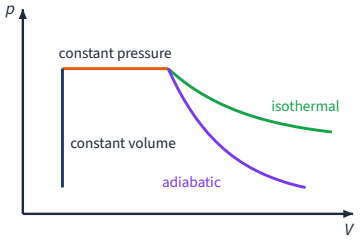
$$C_V = \tfrac{f}{2}R \qquad C_p = C_V + R$$

$$\gamma = \frac{C_p}{C_V}$$

Gas	C_V	C_p	γ
Monatomic	$\frac{3}{2}R$	$\frac{5}{2}R$	5/3
Diatomic	$\frac{5}{2}R$	$\frac{7}{2}R$	7/5

$C_p > C_V$: extra heat does work as the gas expands.

5 THE FOUR PROCESSES



Process	Condition	First law
Isothermal	$\Delta T = 0$	$Q = W$
Adiabatic	$Q = 0$	$\Delta U = -W$
Constant pressure (isobaric)	$\Delta p = 0$	$Q = \Delta U + p\Delta V$
Constant volume (isochoric)	$\Delta V = 0$	$Q = \Delta U$

6 ISOTHERMAL VS ADIABATIC

ISOTHERMAL

$$pV = \text{constant}$$

$$W = nRT\ln\frac{V_2}{V_1}$$

ADIABATIC

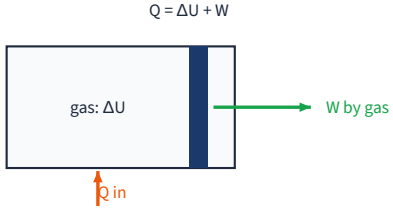
$$pV^{\gamma} = \text{constant}$$

$$TV^{\gamma-1} = \text{constant}$$

$$W = \frac{p_1V_1 - p_2V_2}{\gamma - 1}$$

Adiabatic curve is **steeper** ($\gamma > 1$); adiabatic expansion cools the gas.

7 ENERGY FLOW PICTURE



Isothermal: slow, good thermal contact. Adiabatic: fast, insulated.

8 COMMON SECTION A TRAPS

- Writing $\Delta U = Q + W$ with W done **by** the gas.
- Using $nC_p\Delta T$ for ΔU; it is always $nC_V\Delta T$.
- Taking $Q = 0$ as $\Delta T = 0$ (that is isothermal).
- Forgetting kPa → Pa or L → m³ in $p\Delta V$.
- Reading the area enclosed with the wrong sign for an anticlockwise cycle.

EXAM TIPS

- ✓ Make a table of Q, W, ΔU for each stage of a cycle.
- ✓ Check: around a cycle $\sum \Delta U = 0$.
- ✓ Find the missing temperature with $pV = nRT$.
- ✓ State the sign convention you use.